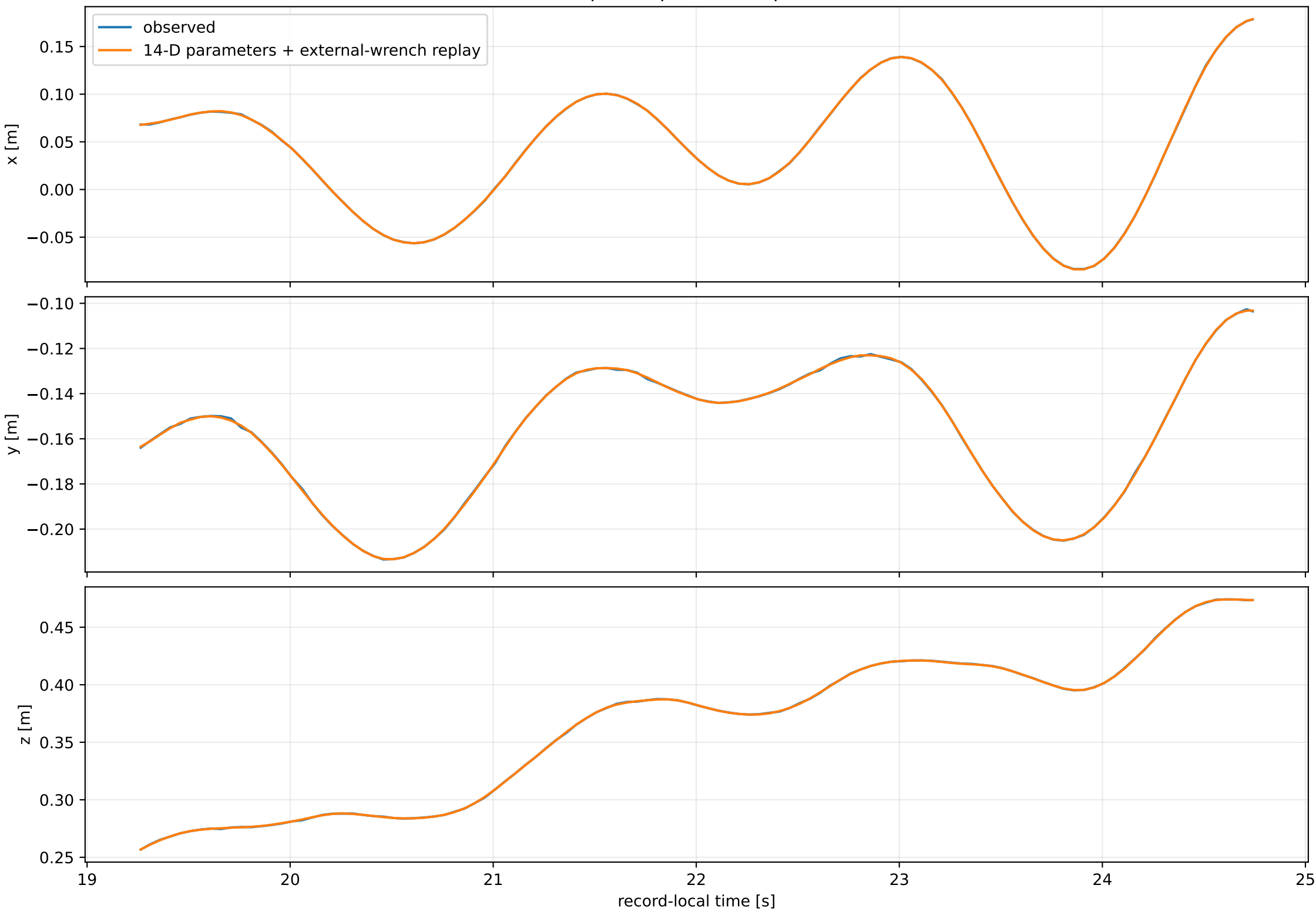
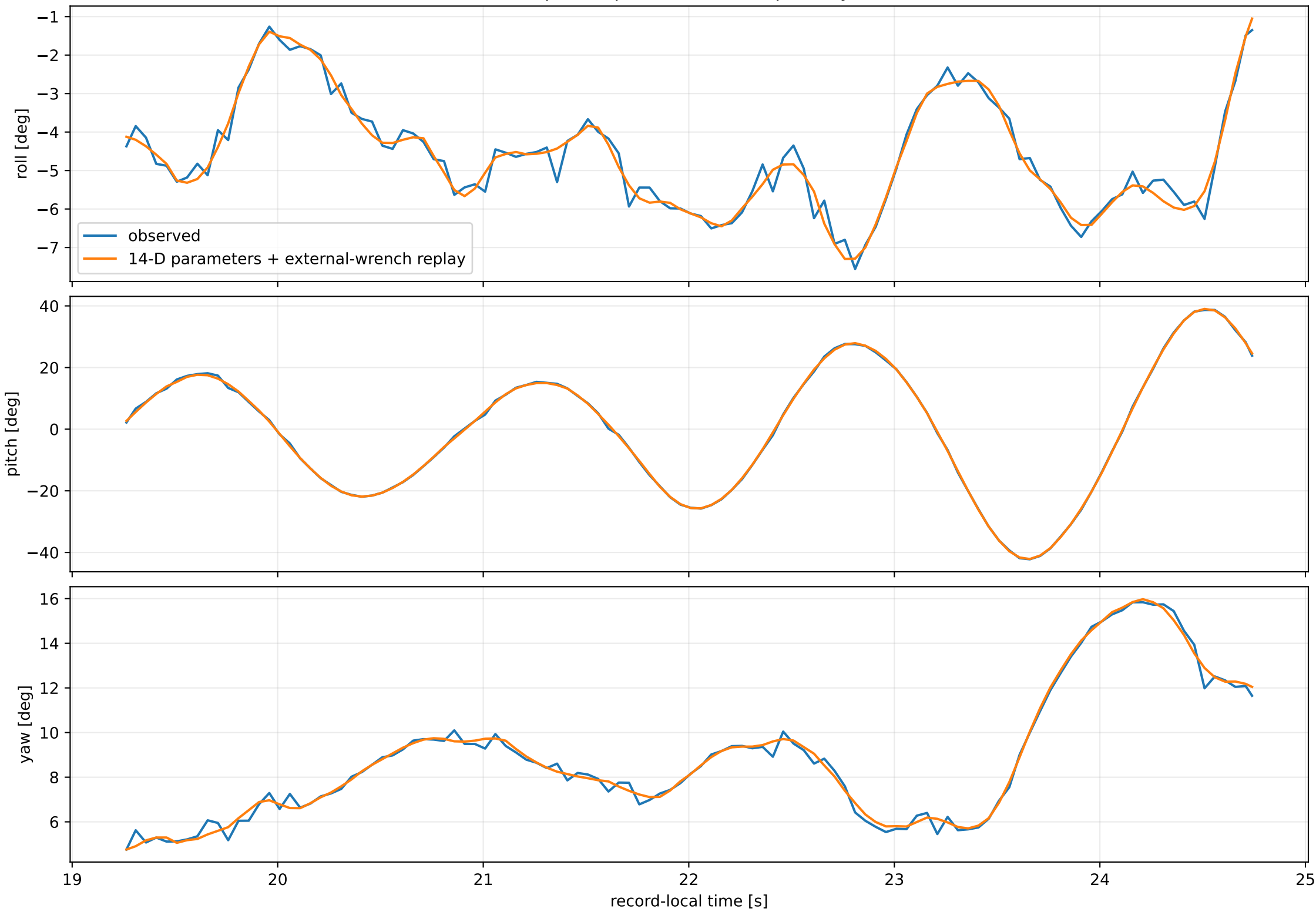


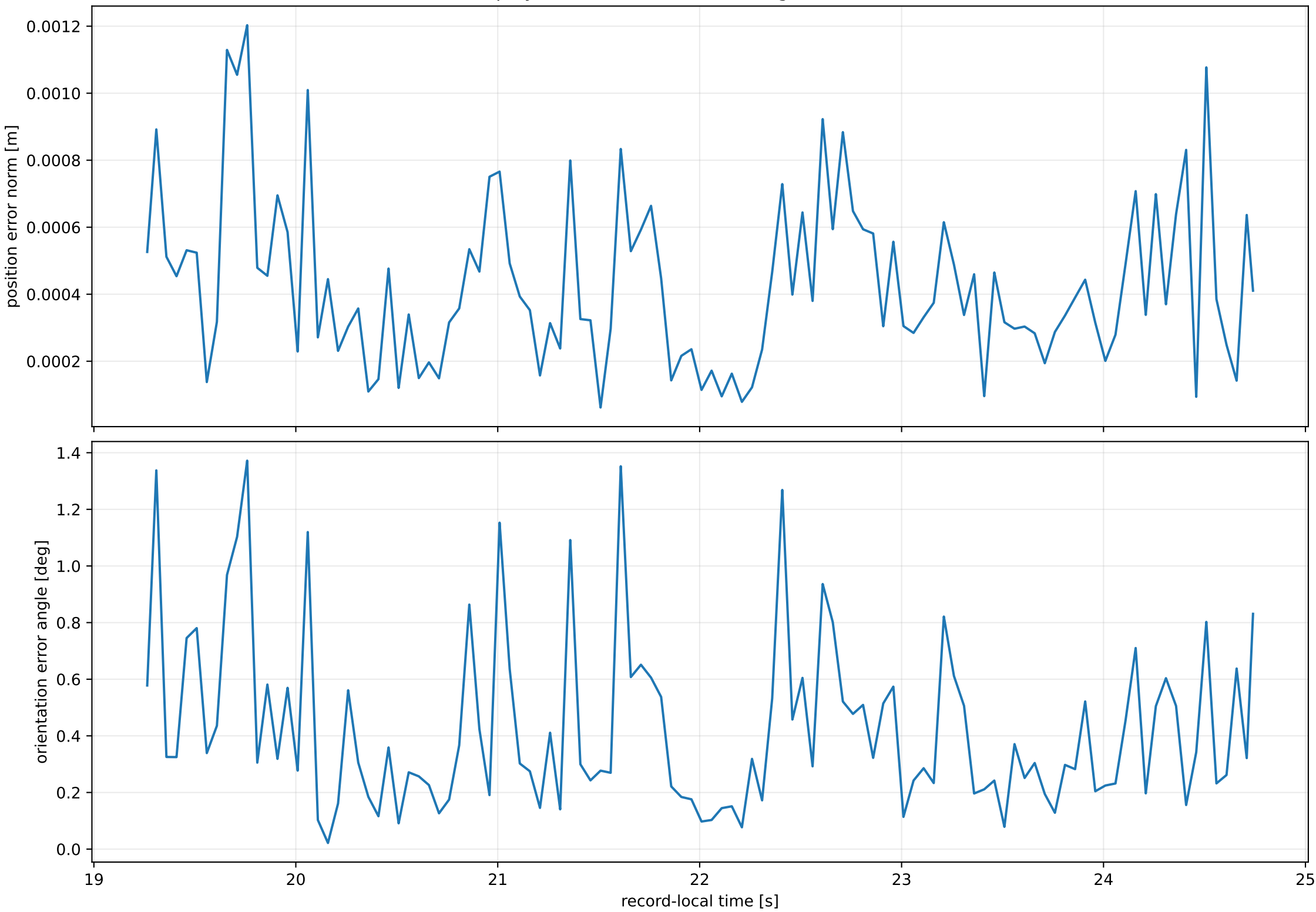
Fine pose reproduction: position



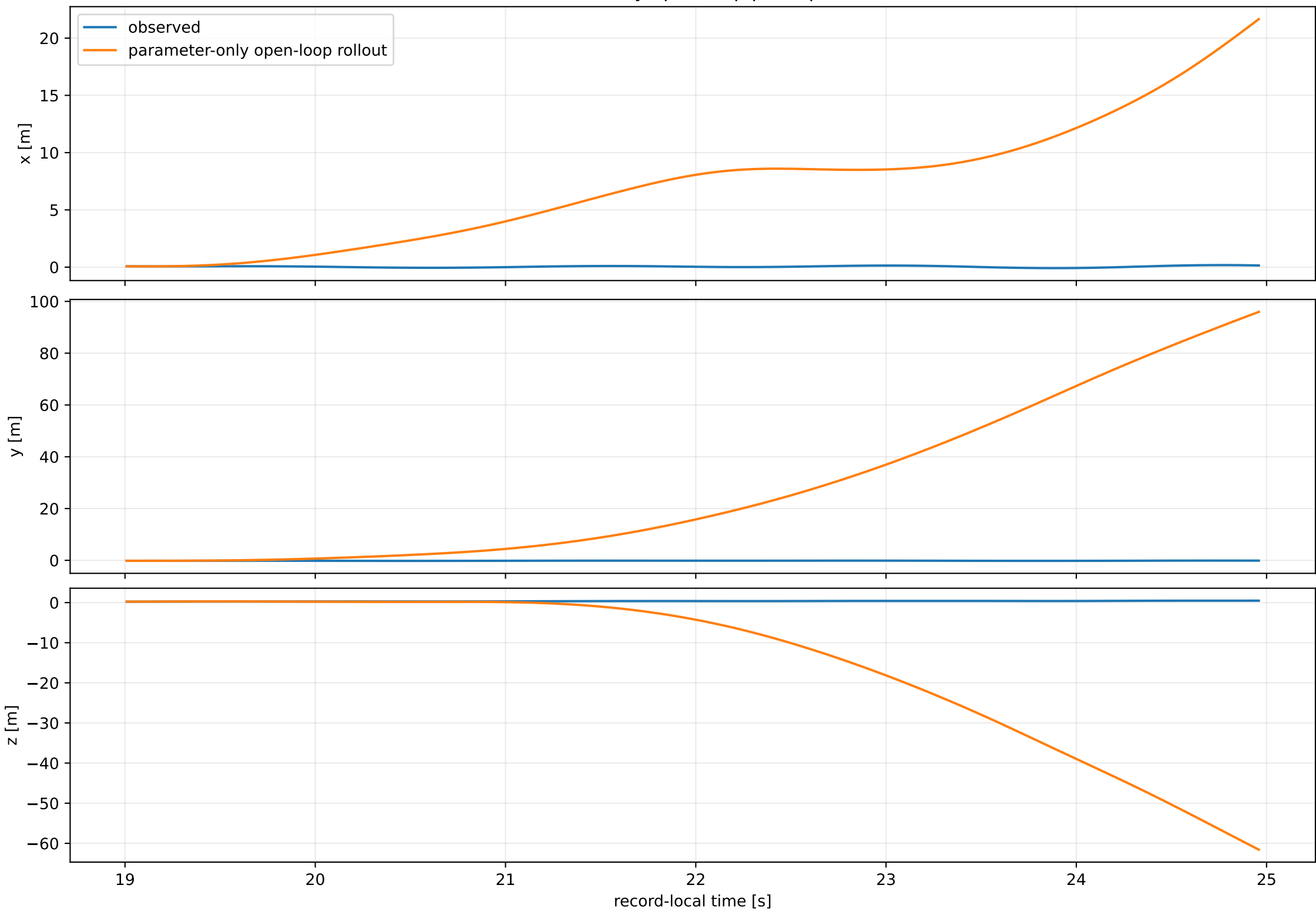
Fine pose reproduction: roll / pitch / yaw



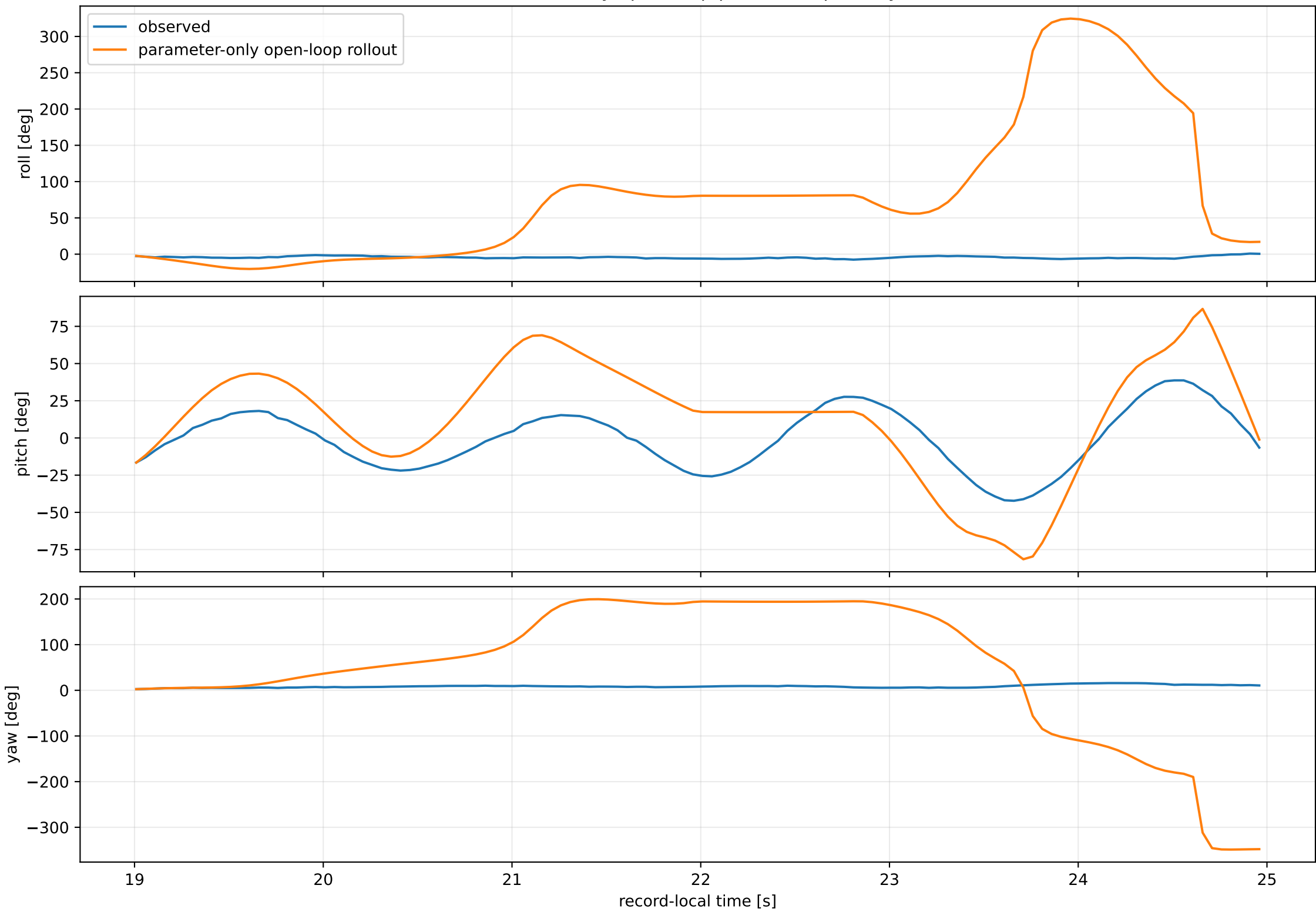
Fine replay error after the 14-D change — lower is better



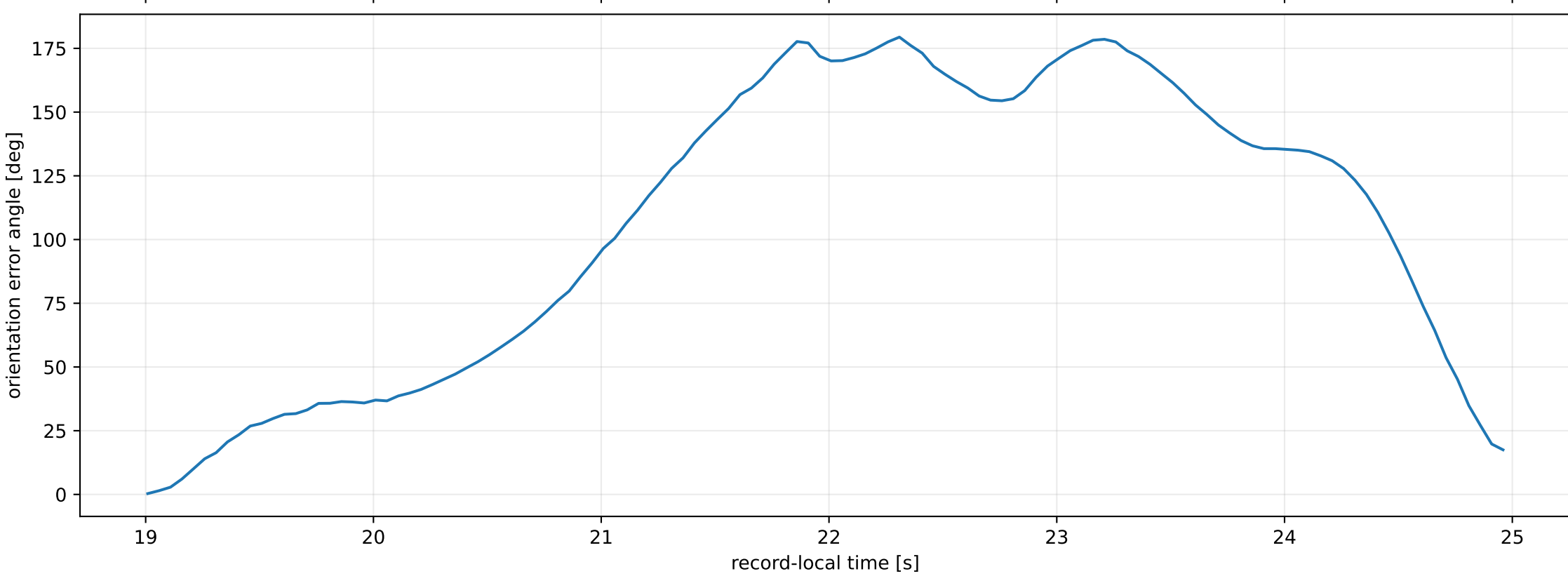
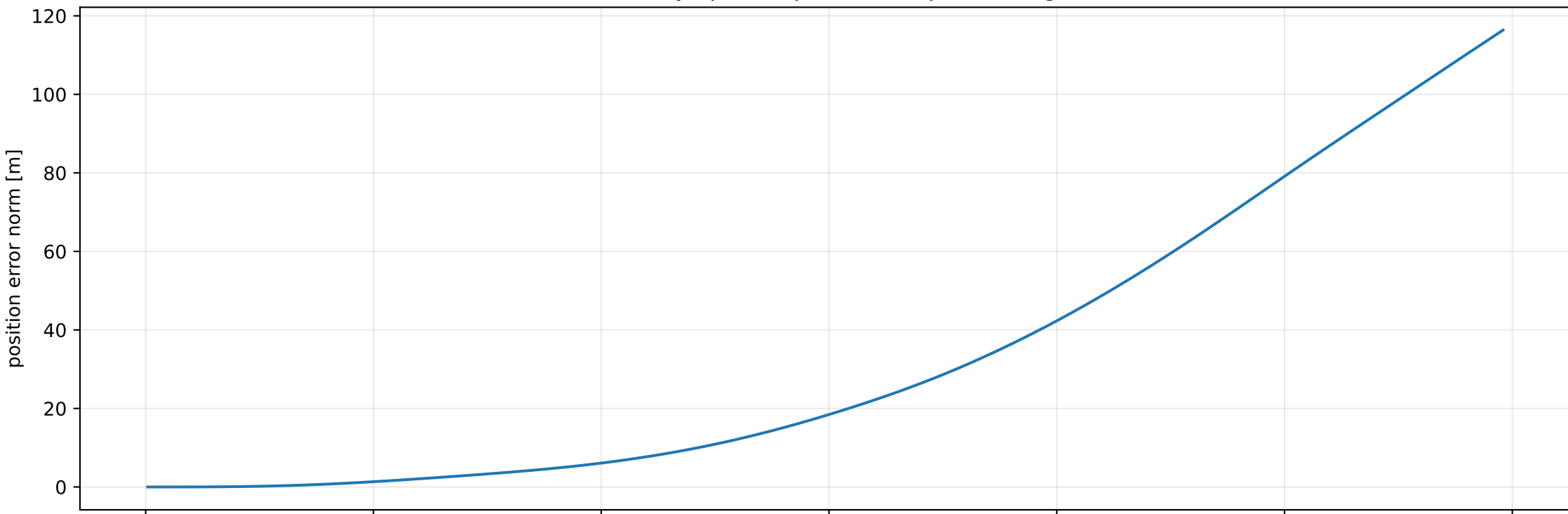
Parameter-only open-loop pose: position



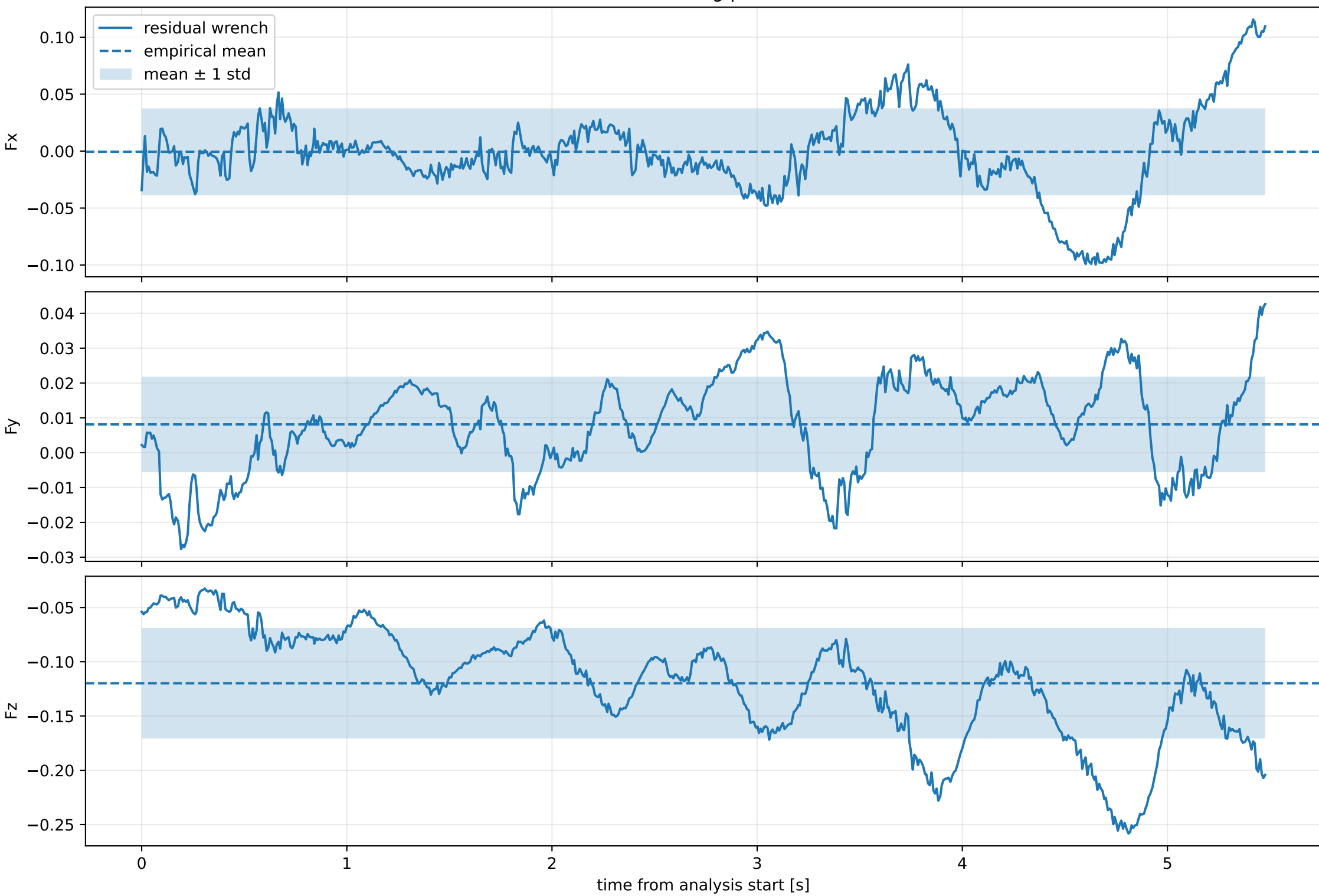
Parameter-only open-loop pose: roll / pitch / yaw



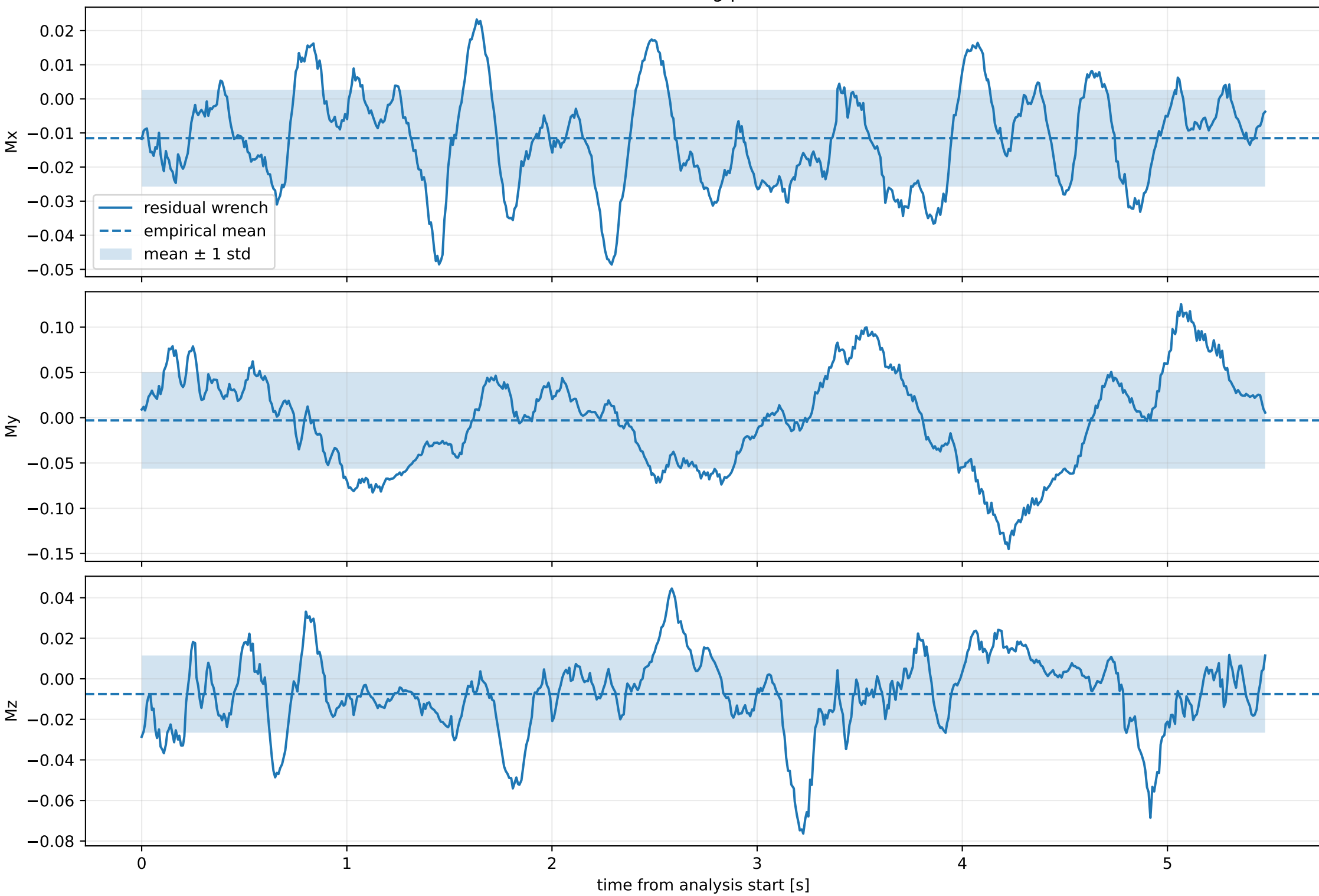
Parameter-only open-loop rollout (separate diagnostic)



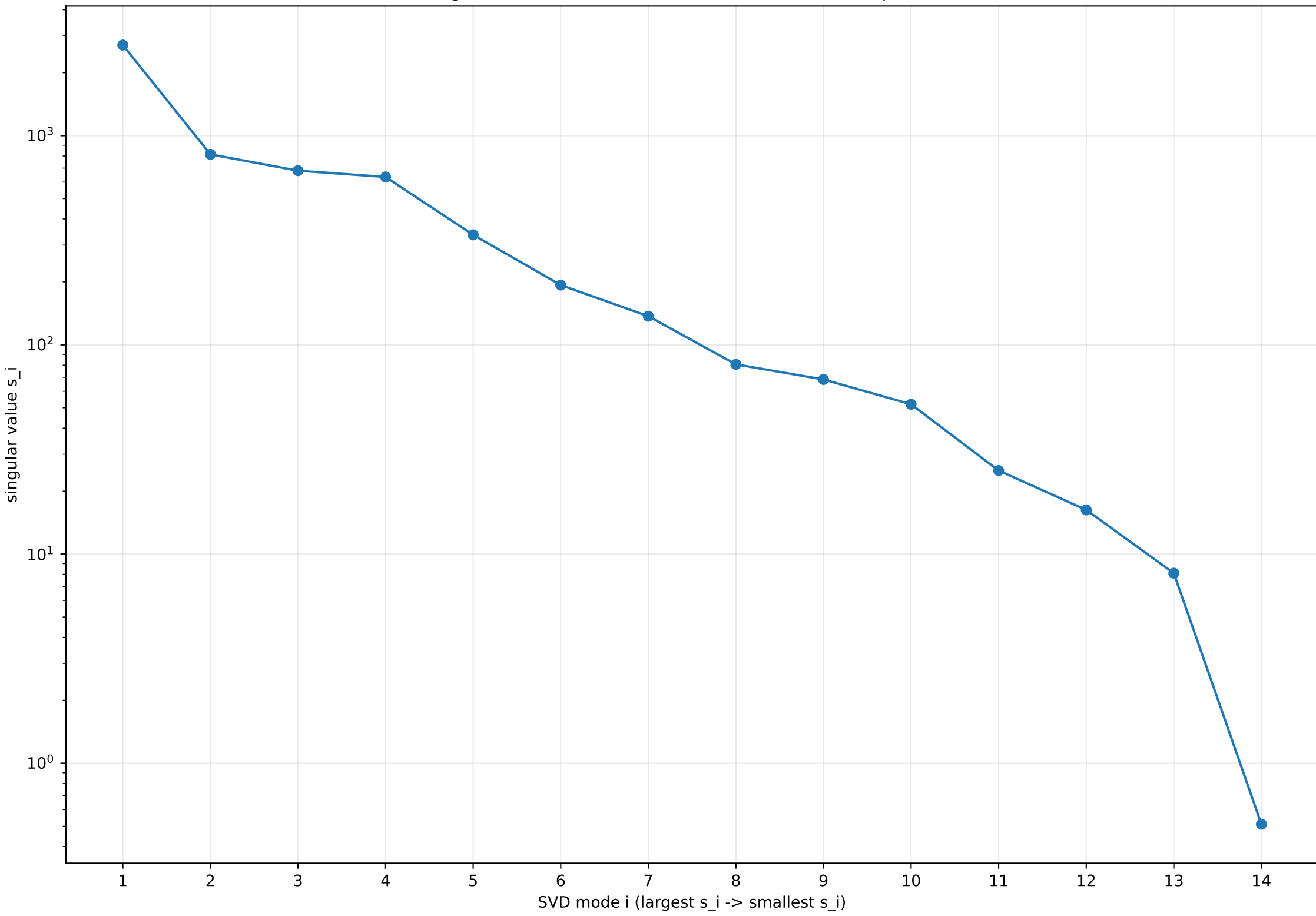
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



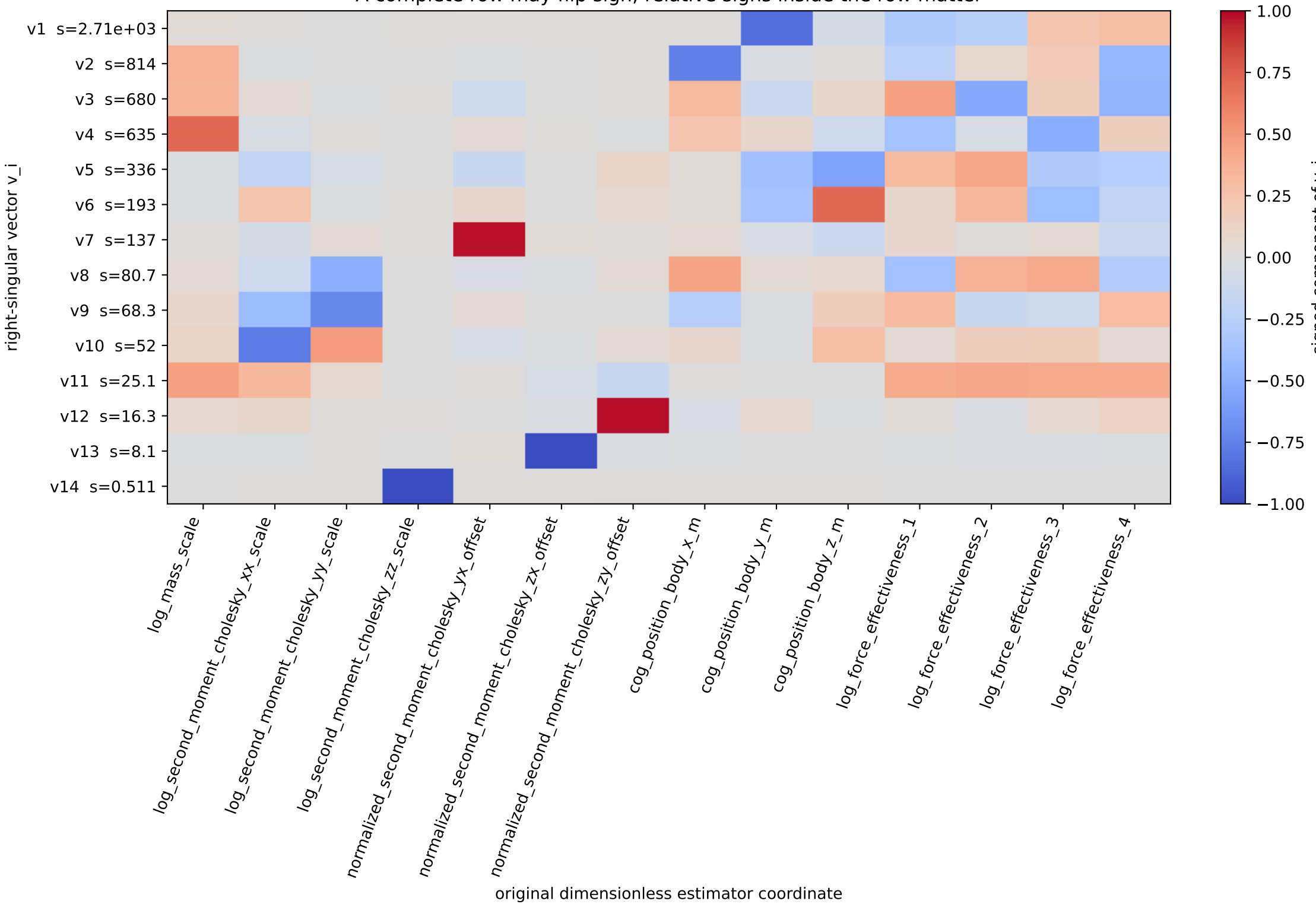
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36576207949

inertia J [kg m²]

[0.065010107 -2.3715287e-05 2.3002863e-05]
[-2.3715287e-05 0.065244869 5.4964758e-06]
[2.3002863e-05 5.4964758e-06 0.12899374]

principal moments [kg m²]

[0.065007726, 0.065247241, 0.12899375]

CoG position in body frame [m]

[-0.0021882396, 0.00086049971, 0.010541833]

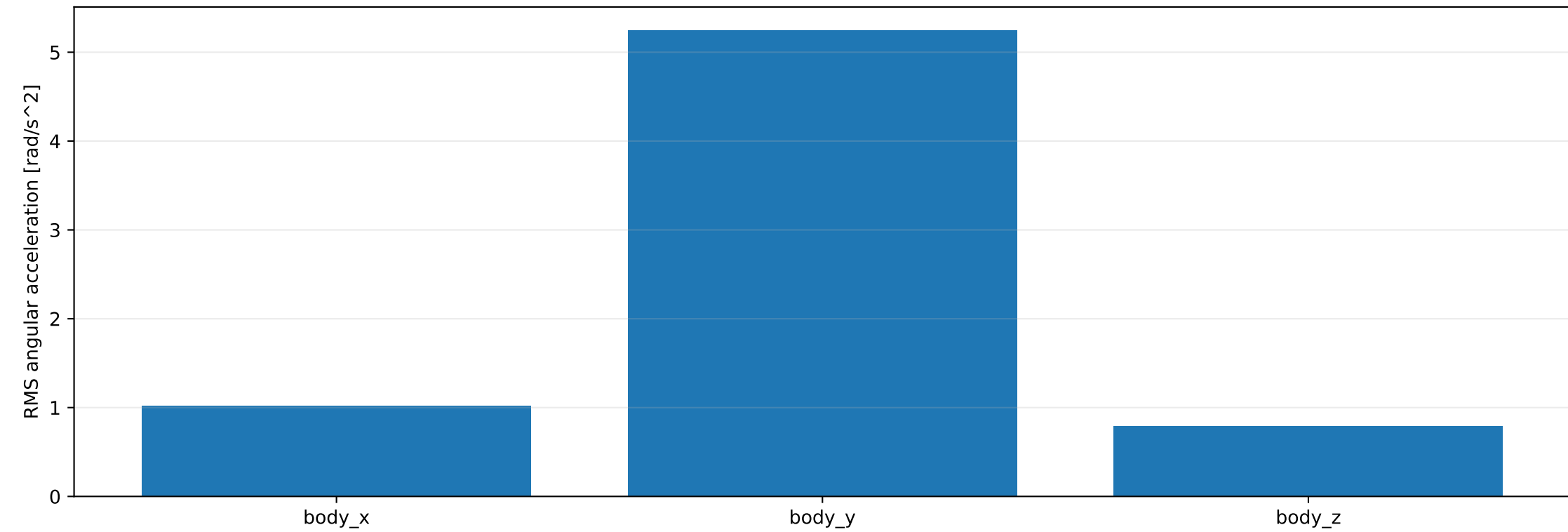
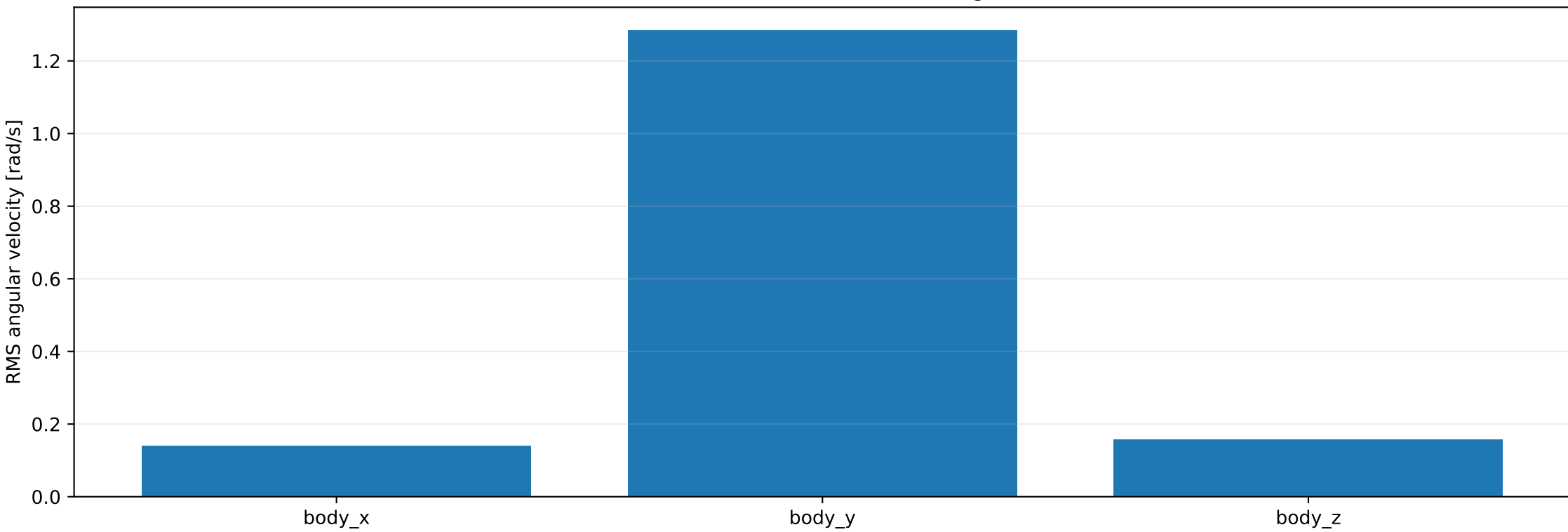
rotor force effectiveness

[0.96694077, 0.97194169, 0.95747781, 0.95600509]

selected delay [s]

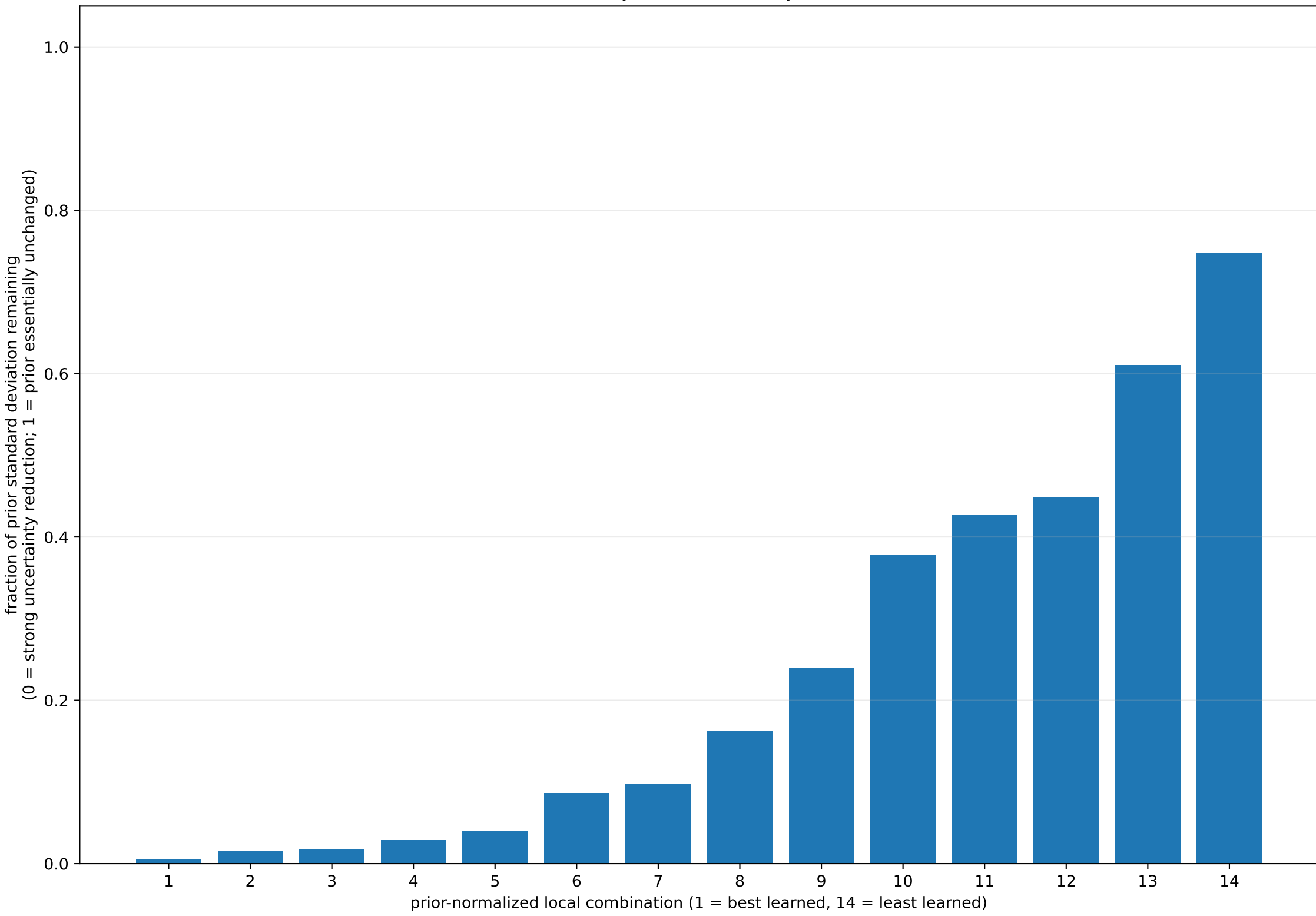
0.2

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

